“Simulated SIEM logs based on real-world attack patterns”

Incident Report: Brute Force Attack Detection

Alert Title

Multiple Failed Logins Followed by Successful Authentication

Summary

A suspicious authentication pattern was detected involving multiple failed login attempts followed by a successful login for the user `usman.shahid`. The activity originated from an external IP address not previously associated with the user. The login behavior deviates from the user’s normal geographic and device patterns, indicating a potential brute force attack leading to account compromise.

Initial Findings

- User: `usman.shahid`
- Source IP: `103.45.67.89`
- Failed Login Attempts: 25
- Successful Login: Yes
- Time Window: 8:00 PM – 8:07 PM
- Location: Singapore (Unusual)
- Normal Location: Pakistan
- Device: Unknown

Key observations:

- High number of failed login attempts in a short period
- Successful login immediately after failed attempts
- Login from a foreign location
- New/unrecognized device

Investigation

Failed Login Analysis

```
SecurityEvent
| where EventID == 4625
| where Account == "usman.shahid"
| summarize FailedAttempts = count() by IPAddress, bin(TimeGenerated, 10m)
| where FailedAttempts >= 20
```

Result: Confirmed 25 failed login attempts from IP `103.45.67.89` within a short time window.

Successful Login Correlation

```
let Failed = SecurityEvent
| where EventID == 4625
| where Account == "usman.shahid"
| summarize FailedCount = count(), LastFail = max(TimeGenerated) by
IPAddress;

let Success = SecurityEvent
| where EventID == 4624
| where Account == "usman.shahid"
| project IPAddress, SuccessTime = TimeGenerated;

Failed
| join kind=inner (Success) on IPAddress
| where SuccessTime between (LastFail .. LastFail + 10m)
```

Result: Successful login occurred shortly after failed attempts from the same IP.

Behavioral Analysis

- Login occurred from an unusual geographic location (Singapore)
- User’s baseline location is Pakistan
- Login was performed from an unrecognized device

Threat Intelligence Check

- IP reputation checked via external intelligence sources
- IP associated with suspicious/malicious activity patterns

Evidence

- 25 failed login events (Event ID 4625)
- 1 successful login event (Event ID 4624)
- Same IP used for all attempts
- Unusual login location
- New device fingerprint

Conclusion

This incident is classified as a **True Positive**.
The observed pattern strongly indicates a brute force attack that successfully compromised the user account.

Response Actions

- User account temporarily disabled
- Password reset enforced
- Active sessions revoked
- Incident escalated to Tier 2 for further investigation
- Monitoring enabled for related activity

Severity Level

High

Analyst Note

The attack demonstrates a classic brute force pattern followed by successful authentication. Immediate containment prevented further misuse of the compromised account. Additional monitoring is recommended to ensure no lateral movement or further malicious activity occurs.